ABRA — Project Summary

Version 2.4.0 · 2026-07-23 · Will Hooper

A one-page map of the whole project and every component. For depth: the white paper (math + sources), the deck (plain-English), the technical docs (how to run it), and the living model ledger.

What ABRA is

ABRA is a decision-support model family for **Pokémon Champions VGC, Reg M-B, best-of-one closed-sheet ladder**. It stores every public ladder replay and builds small, CPU-trainable models on that growing store. It runs in a browser with no build step.

The finding that shapes everything

You cannot reliably predict who wins from the two team sheets — even a player-rating model ties a coin. So ABRA does not sell outcome prediction. It supports *decisions* and grades every model with a proper score, a confidence interval, and an honest baseline. Wins are reported as wins; two honest negatives are reported as negatives.

The components at a glance

Model	What it is	Status	Headline result
MEDICHAM	Exact Gen-9 doubles damage engine	□Validated	Within 5% of `@smogon/calc` on 100% of 31 scenarios (median 0%)
GURU	Meta matchup matrix from real outcomes	□Built	13 archetypes over 5,199 games, Wilson CIs (descriptive; winner-prediction ties a coin, as expected)
XATU	Opponent set + next-move belief	□Built	Top-1 36% / top-3 72% on held-out human moves (beats its baselines)
PORY	Mid-game win-probability value net	□**The win**	Log-loss 0.567 vs coin 0.693, calibrated (ECE 1.6%); live in KADABRA
CHOMP	Bring-4 / lead-2 team-preview engine	□Ships (standalone)	Exact-damage picker; **CHOMP-EV proof: brings tie a coin (honest null)**
SLOWKING	Team-preview Nash (mixed strategy)	□Built	Equilibrium ≪ exploitable than uniform; playstyle cycle is **suggestive on small samples**
KADABRA	Replay coach	□Works offline	Per-turn "you're at X%" from PORY
DITTO	Team optimiser	⚠ Pivoting	Objective de-biased to validated damage (was optimising a backwards signal)
ALAKAZAM	In-battle decision engine (capstone)	□In development	Belief + search + learned value; built last on the inputs above

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MEW	Self-play data engine	□Roadmap	The path to the millions of games ALAKAZAM's strongest version needs

How it fits together

The **store** (every real game) feeds **GURU** (meta), **XATU** (belief), **PORY** (value), and **MEDICHAM** (damage). **SLOWKING** solves the preview; **CHOMP** picks the bring; **KADABRA** coaches a replay with PORY. **ALAKAZAM** is the capstone that assembles belief + search + value into the win-%-optimal move, built last. Every change updates the code, this summary, the white paper, the deck, the technical docs, and the CHANGELOG in the same pass.

Repositories and site

Piece	Repo	Live
ABRA (models + site)	`github.com/willhoop/abra`	`willhoop.github.io/abra/app/`
CHOMP (bring engine)	`github.com/willhoop/chomp`	Showdown userscript
Portfolio	`github.com/willhoop/willhoop.github.io`	`willhoop.github.io`

Honest ceilings

Predicting the match winner from sheets is a permanent coin flip in this format. The meta-structure models (playstyle, cores) are small-sample and stay suggestive until the store grows (~18k games/week). The load-bearing wins are the validated damage and the mid-game value net.